WORKED SOLUTION

STEP II 2017, Question 1

Pure • Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q1

The key idea

The arctan x is the awkward part of the integrand, but its derivative is a rational function, so integration by parts with $u = \arctan x$ swaps it for $\frac{1}{1+x^2}$. Then the trick in (ii) is to pair two integrals so that the numerators combine to $x^{n+1}(1+x^2)$ and the troublesome denominator cancels. That gives a reduction formula stepping n by 2; using it twice steps n by 4, which is exactly what the induction in (iii) needs.

Solution

Part (i)

Integrate by parts with $u = \arctan x$ and $\mathrm{d}v = x^n\,\mathrm{d}x$, so $v = \frac{x^{n+1}}{n+1}$ (fine, since $n \geq 0$):

$$\begin{aligned} I_n &= \left[\frac{x^{n+1}}{n+1}\arctan x\right]_0^1 - \int_0^1 \frac{x^{n+1}}{n+1}\cdot\frac{1}{1+x^2}\,\mathrm{d}x \\ &= \frac{1}{n+1}\cdot\frac{\pi}{4} - \frac{1}{n+1}\int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x, \end{aligned}$$

using $\arctan 1 = \frac{\pi}{4}$ and the fact that the boundary term vanishes at $x = 0$. Multiplying by $n+1$ gives the required result:

$$(n+1)I_n = \frac{\pi}{4} - \int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x.$$

For $n = 0$ the remaining integral is $\int_0^1 \frac{x}{1+x^2}\,\mathrm{d}x = \left[\frac{1}{2}\ln(1+x^2)\right]_0^1 = \frac{1}{2}\ln 2$, so

$$I_0 = \frac{\pi}{4} - \frac{1}{2}\ln 2.$$

Part (ii)

Apply (i) with $n+2$ in place of n and add it to (i) as it stands:

$$\begin{aligned}(n+3)I_{n+2}+(n+1)I_n &= \frac{\pi}{2}-\int_0^1 \frac{x^{n+3}+x^{n+1}}{1+x^2}\,\mathrm{d}x\\ &= \frac{\pi}{2}-\int_0^1 x^{n+1}\,\mathrm{d}x\\ &= \frac{\pi}{2}-\frac{1}{n+2},\end{aligned}$$

because $x^{n+3}+x^{n+1}=x^{n+1}(1+x^2)$.

Ask yourself The formula links I_4 to I_2, and I_2 to I_0. Which values of n do we need?

We need $n=0$ and $n=2$:

$$\begin{aligned}n=0:&\quad 3I_2+I_0=\frac{\pi}{2}-\frac{1}{2},\\ n=2:&\quad 5I_4+3I_2=\frac{\pi}{2}-\frac{1}{4}.\end{aligned}$$

Subtracting the first from the second eliminates I_2:

$$5I_4=I_0+\frac{1}{4}=\frac{\pi}{4}+\frac{1}{4}-\frac{1}{2}\ln 2,\qquad\text{so}\qquad I_4=\frac{\pi+1-2\ln 2}{20}.$$

Part (iii)

Write $S_n=\sum_{r=1}^{2n}\frac{(-1)^r}{r}$ for short.

Finding A. For $n=1$, $S_1=-1+\frac{1}{2}=-\frac{1}{2}$, so the claim reads $5I_4=A+\frac{1}{4}$. From (ii), $5I_4=\frac{\pi}{4}+\frac{1}{4}-\frac{1}{2}\ln 2$, so we must take

$$A=\frac{\pi}{4}-\frac{1}{2}\ln 2\ (=I_0),$$

and with this value the claim is true for $n=1$.

Inductive step. Suppose $(4k+1)I_{4k}=A-\frac{1}{2}S_k$ for some $k\geq 1$. We want to reach I_{4k+4}, which is two steps of the formula from (ii). With $n=4k+2$ and $n=4k$ it gives

$$\begin{aligned}(4k+5)I_{4k+4}+(4k+3)I_{4k+2}&=\frac{\pi}{2}-\frac{1}{4k+4},\\ (4k+3)I_{4k+2}+(4k+1)I_{4k}&=\frac{\pi}{2}-\frac{1}{4k+2}.\end{aligned}$$

Ask yourself How do we get rid of the unwanted I_{4k+2}?

It appears with the same coefficient in both equations, so subtracting the second from the first removes it:

$$(4k+5)I_{4k+4} = (4k+1)I_{4k} + \frac{1}{4k+2} - \frac{1}{4k+4}.$$

On the other side, S_{k+1} has two more terms than S_k, namely $r = 2k+1$ (odd, so negative) and $r = 2k+2$ (even, so positive):

$$-\frac{1}{2}S_{k+1} = -\frac{1}{2}S_k - \frac{1}{2}\Big(-\frac{1}{2k+1} + \frac{1}{2k+2}\Big) = -\frac{1}{2}S_k + \frac{1}{4k+2} - \frac{1}{4k+4}.$$

Using the inductive hypothesis,

$$(4k+5)I_{4k+4} = A - \frac{1}{2}S_k + \frac{1}{4k+2} - \frac{1}{4k+4} = A - \frac{1}{2}S_{k+1},$$

which is the claim for $n = k+1$. Since it holds for $n = 1$, by induction it holds for all $n \geq 1$, with $A = \frac{\pi}{4} - \frac{1}{2}\ln 2$.

Checks

- Numerically, $I_0 \approx 0.438825$ and $I_4 \approx 0.137765$; Simpson's rule on the original integrals agrees to ten decimal places.
- A equals I_0, so the formula in (iii) is also true for $n = 0$ (the sum is then empty), a good sign.
- As $n \to \infty$, the integral in (i) tends to 0 (the integrand is at most x^{n+1}), so $(4n+1)I_{4n} \to \frac{\pi}{4}$. Meanwhile $S_n \to -\ln 2$, so the right-hand side tends to $\frac{\pi}{4} - \frac{1}{2}\ln 2 + \frac{1}{2}\ln 2 = \frac{\pi}{4}$. The two sides agree in the limit.

Common slips

- Writing $v = \frac{x^{n+1}}{n+1}$ but forgetting to divide the second integral by $n+1$ too.
- Getting the signs of the two new terms of the alternating sum wrong: $r = 2k+1$ is odd, so $(-1)^r = -1$.
- Proving the inductive step without first fixing A from the base case.

Beyond STEP

The final check uses the alternating harmonic series $1 - \frac{1}{2} + \frac{1}{3} - \cdots = \ln 2$, and this question in effect proves it: the identity in (iii) plus the fact that $(4n+1)I_{4n} \to \frac{\pi}{4}$ forces the partial sums to converge to $\ln 2$. Working with integrals like $\int_0^1 \frac{x^{n+1}}{1+x^2}\,\mathrm{d}x$ that shrink to zero is a standard way to control the error in a series, and you will meet it in a first analysis course. Abbott's *Understanding Analysis* is a friendly place to see how such limits are made rigorous.

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